

Aya Alaoui Sossé

aya.alaouisossee@gmail.com • +212 650 594 635 • linkedin.com/in/aya-alaoui-sosse • github.com/aya-alaoui-sossee

AI Researcher — Computational Pathology & Trustworthy AI Biomarkers in Oncology

Research Interests

Trustworthy AI quantification of tumour-infiltrating lymphocytes (TILs) from whole-slide images; immune biomarkers in triple-negative breast cancer; fairness and generalization of computational-pathology models across populations, especially underrepresented African / Moroccan cohorts; and molecular foundation models for digital pathology (incl. EGFR mutation prediction).

Education

Ph.D. in Artificial Intelligence & Data Science

2025 – Present

Mohammed V University, Faculty of Science — Rabat, Morocco

Thesis: AI-Driven Molecular Foundation Model for Cancer Pathology and Precision Medicine. *Research areas:* computational pathology, immune biomarkers (tumour-infiltrating lymphocytes) in breast cancer, and trustworthy AI across patient populations.

M.Eng. in Big Data & Artificial Intelligence

2022 – 2025

Euromed University of Fez (UEMF) — Morocco. Fully funded — full (100%) merit scholarship for the engineering programme.

Relevant coursework: Deep Learning, Medical Image Analysis, Statistical Modelling, Large-Scale Data Systems.

Training & Selected Programmes

CCAIM Machine Learning for Healthcare Summer School — University of Cambridge

2026

Accepted for a **fully funded place supported by GSK**. Presented in the exhibition session: “*Do immune biomarkers stay true across populations? Trustworthy AI for TILs in triple-negative breast cancer.*”

Research Experience

Doctoral Researcher — Trustworthy TIL Biomarkers

2025 – Present

Mohammed V University · DataPathology — Morocco

- Developing AI methods to quantify tumour-infiltrating lymphocytes (TILs) from H&E whole-slide images in triple-negative breast cancer, with a focus on whether the biomarker stays reliable across different patient populations and imaging sources.

AI & Data Scientist — Medical AI

2026 – Present

DataPathology — Remote / Morocco

- Developing high-accuracy AI models to predict mutations from whole-slide images (94% benchmark accuracy), as part of applied computational-pathology and precision-oncology research.

R&D Intern — Medical AI, DataPathology

Feb – Aug 2025

Engineered an end-to-end EGFR prediction pipeline from whole-slide images, outperforming existing solutions by 15% on benchmark datasets.

Undergraduate Researcher

2022 – 2025

Euromed University of Fez (UEMF) — Morocco

- **TIL Spatial Quantification & Tumour Subtype Classification** (2024–2025): multi-task foundation model for joint TIL quantification and adenocarcinoma / adenoma classification for precision immunotherapy.
- **Breast Cancer Detection AI** (2023–2024): computer-vision system, 94% accuracy, 60% faster diagnosis, validated against physician annotations.

Publications

Alaoui Sossé, A. et al. *A Systematic Mapping Study of Deep Learning for EGFR Mutation Prediction in Lung Adenocarcinoma*. Book chapter, proceedings of the Int. Conference on Modern Artificial Intelligence & Data Science (MAIDS 2026), Mohammed V University, Rabat. **Accepted.**

Languages

Arabic — Native • French — C2 Professional • English — C2 Professional

Volunteering & Outreach

Conference Organiser & Representative, DataPathology (2025 – Present) — Co-organised the 1st Congress of Digital Pathology and represented DataPathology at 2 further international AI-in-oncology conferences.

English Teacher & Camp Facilitator, Youth Education Outreach (Mar 2023 – Feb 2025) — Taught English, algorithms and logic to young learners through summer camps and workshops.